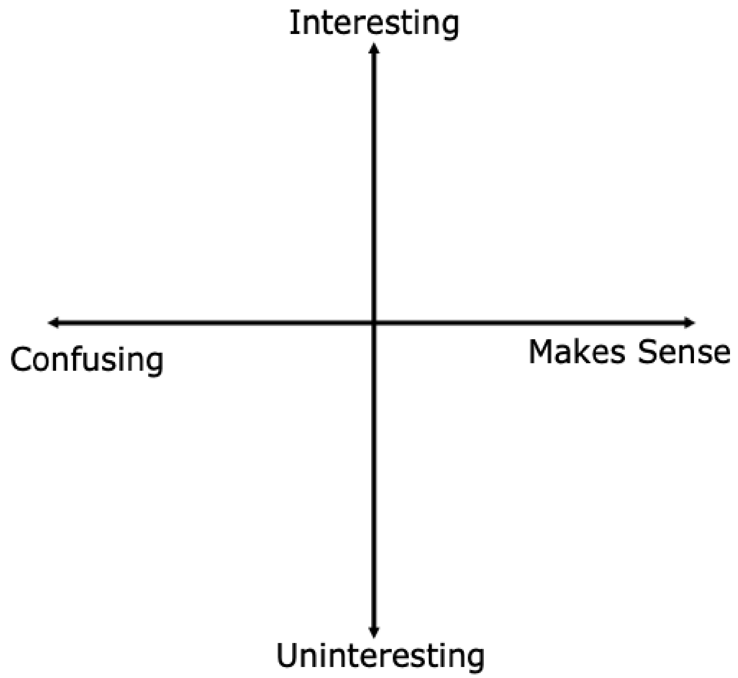
2.7.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's lesson.



Unit 2, Lesson 8: Proofs – Part 2

Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

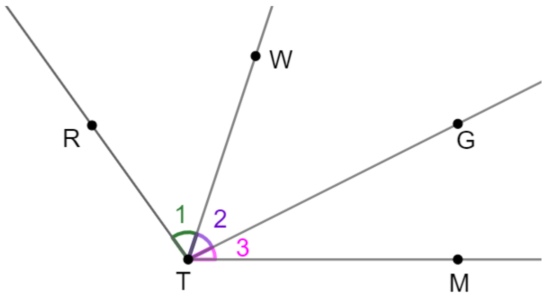
MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

Learning Targets

- ☐ I can reflect on the mathematical thinking found in a proof.
- ☐ I can reason about proofs using deductive reasoning, definitions, and postulates.

2.8.1 Warm-Up: Bell Work

Given: $m\angle 1 = m\angle 3$
Prove: $\angle RTG \cong \angle WTM$



1. Use the flowchart proof and the two-column proof from the previous lesson to write a paragraph proof.

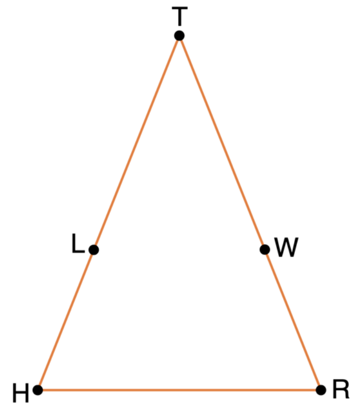
2.8.2 Collaboration: Brainstorming Before Writing a Proof

Work with your partner to complete the following.

Gabrielle and Samuel were given the following information.

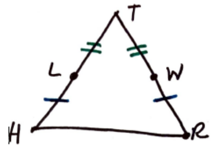
ΔTHR is shown where $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$.

Prove that $\overline{HT} \cong \overline{TR}$.



Before Gabrielle and Samuel wrote their paragraph proof, they brainstormed. Their notes are shown below.

Gabrielle



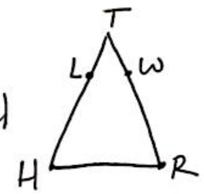
Given: $\overline{HL} \cong \overline{WR}$ $\overline{LT} \cong \overline{WT}$ Prove: $\overline{HT} \cong \overline{TR}$

$HT = HL + LT$ adding two segments to get the total

What is that called?

If using = then it has to be changed to congruence.

Samuel



Given: $\overline{HL} \cong \overline{WR}$ $\overline{LT} \cong \overline{WT}$

Prove: $\overline{HT} \cong \overline{TR}$ don't worry, it's the same as $\overline{RT}$

Basically if you add the two segments of each side you get the entire side.

1. Why did Samuel write "Don't worry it is the same thing as $\overline{RT}$?"

2. Gabrielle made some markings on the triangle she drew. What are her markings meant to show?

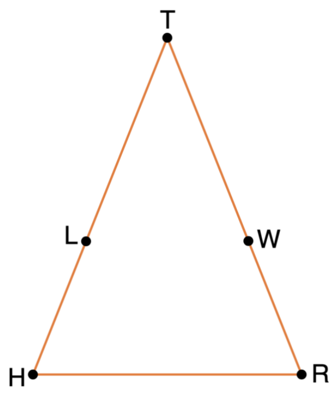
3. Gabrielle wrote "$HT = HL + LT$," but she could not remember what that was called. Samuel wrote "Basically if you add the two segments of each side you get the entire side." What postulate justifies Gabrielle's statements?

   If you and your partner cannot remember the name of the postulate Gabrielle and Samuel are referring to, then look through your workbook to find it.

4. Gabrielle only wrote "$HT = HL + LT$," but Samuel mentioned "each side." Write the equation for the other side of the triangle.

5. Gabrielle wrote "If using = then it has to be changed to congruence." What definition justifies changing from equality to congruence?

6. Apply the brainstorming of Gabrielle and Samuel to finish the paragraph proof. The first sentence has been written for you.



Given: $\overline{HL} \cong \overline{WR}$, $\overline{LT} \cong \overline{WT}$
Prove: $\overline{HT} \cong \overline{TR}$

It is given that $\overline{HL} \cong \overline{WR}$ and $\overline{LT} \cong \overline{WT}$.

7. Describe the most difficult part of writing the paragraph proof.

8. Describe how Gabrielle and Samuel’s brainstorming was helpful.

9. Check your proof with another group and if necessary, resolve any differences.

2.8.3 Guided Instruction: Flowchart Proof

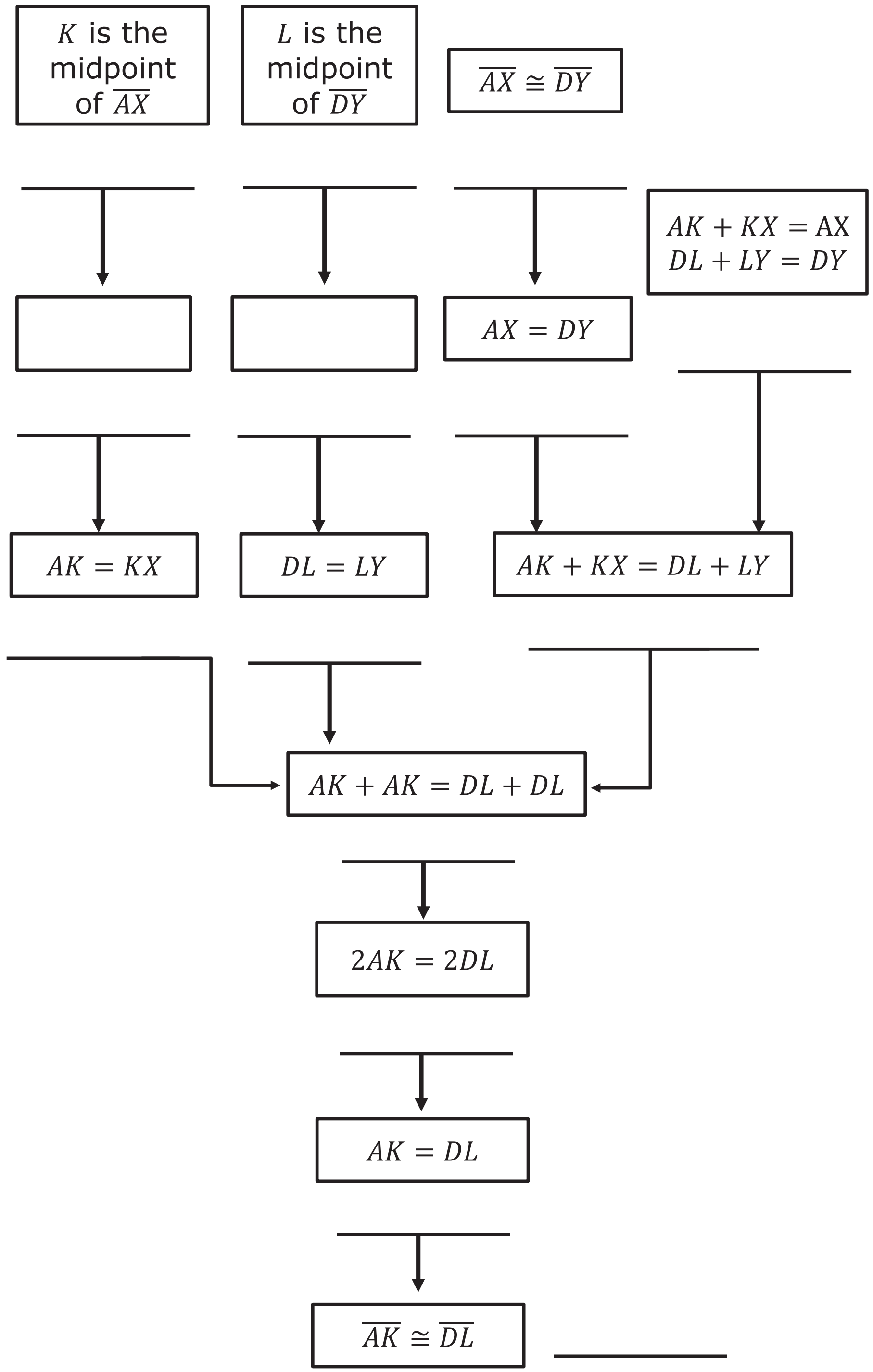
$\overline{AX}$ and $\overline{DY}$ are shown.

Given: K is the midpoint of $\overline{AX}$, L is the midpoint of $\overline{DY}$, and $\overline{AX} \cong \overline{DY}$

Prove: $\overline{AK} \cong \overline{DL}$

1. Brainstorm with your partner about how to write the proof. Then, write your brainstormed ideas in the space provided. Include any possible definitions or postulates that may need to be used to justify the steps.

2. Complete the flowchart proof.



2.8.4 Your Turn!

1. Use the previous flowchart to complete the two-column proof.

Statement	Reason
1. K is the midpoint of $\overline{AX}$ and L is the midpoint of $\overline{DY}$	1. Given
2.	2.
3. $AK = KX$ and $DL = LY$	3.
4. $\overline{AX} \cong \overline{DY}$	4. Given
5.	5. Definition of Congruence
6. $AK + KX = AX$ and $DL + LY = DY$	6.
7. $AK + KX = DL + LY$	7. Substitution property of equality
8. $AK + AK = DL + DL$	8.
9. $2AK = 2DL$	9.
10. $AK = DL$	10.
11. $\overline{AK} \cong \overline{DL}$	11.

2.8.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

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